Project: University Transportation Management System

1. Background

In Algerian universities, student transport is an essential service. However, several problems are observed:

- Bus overload
- Poor distribution of students
- Lack of visibility on schedules
- Lack of tracking of assignments
- Ineffective change and incident management

This project aims to put students in a real situation of designing a database, by mobilizing all the notions studied in the module.

2. Objectives

The objective of this project is to design and implement a platform (website or mobile application) to improve the management of university transport. The system must, **among other things**, offer the following functionalities:

- Manage students and their subscriptions
- Manage buses and their capacities
- Manage transmission lines
- Manage stations and routes
- Plan schedules
- Assign buses to routes
- Assign students to rows
- Track assignment history
- Manage incidents and delays

3. Management rules

The following list presents a **number** of business constraints:

- A student can subscribe to only one line at a time
- One line serves several stations
- A station can belong to several lines
- A bus has a maximum capacity
- A bus is assigned to a line for a given period of time
- A line has several schedules depending on the day

- A student can change the line (history to be kept)
- Bus assignments must be logged
- A journey corresponds to a passage of a bus on a line at a given time

4. Work required

- **Part 1: Analysis**
  - Understand the problem
  - Identify additional constraints
- **Part 2: Conceptual Modeling (DCM)**
  - Propose an Entity-Association Model
- **Part 3: Logical Modeling (MLD)**
  - Transformation of the Entity-Association Model into a Relational Model
  - Definition of Primary and Foreign Keys
- **Part 4: Standardization(Normalisation)**
  - Standardization up to **BCNF**(Boyce-Codd Normal Form)
- **Part 5: Implementation**

Below is a list of **some** queries to implement:

- Number of students per line
- Bus occupancy rate
- Timetables of a given line
- Students without season tickets
- History of a student's assignments
- Buses assigned to a line on a given date
- Busiest lines
- List of trips with delays

5. Deliverables

- Report (Analysis + Entity-Association Model + Relational Model + Normalization)
- Application
- Oral presentation

6. Important notes

- Work in groups of **5 students**
- Deadline : **31/05/2026**

Good luck